

parameter update strategy should consider the possibility of no output change for an insufficient step size, despite the gradient in the value-clamped region stating that an output change was expected. Again, this downside is akin to the gradient interpretation difficulties faced when clamping values and being near the boundary on the inside of the allowable region and stepping outside, in which case an output change was also expected but did not realize when the clamping immediately applies.

### G.3 Clamping in fuzzy logic

#### G.3.1 Real-unit interval

The continuous t-norms of fuzzy logic perform binary operations on the real-unit interval  $[0, 1]$  that represent infinite-valued logic where truth values can be interpreted as ambiguous mixtures between True and False. Clamping the result of fuzzy logic operations like those of Łukasiewicz logic is necessary to ensure truth values remain in the interpretable range of  $[0, 1]$ . The  $\min()$  and  $\max()$  functions clamp the computed values in all Łukasiewicz logic operators, otherwise the result can be outside of the permissible truth value range of  $[0, 1]$ . For the Łukasiewicz conjunction upward pass the clamping then bounds the output and provides the associated clamped gradients as follows.

$$\bigotimes_{i \in I}^{\beta} x_i^{\otimes w_i} = \max(0, \min(1, \beta - \sum_{i \in I} w_i(1 - x_i))), \quad (102)$$

$$\frac{\partial \left( \bigotimes_{i \in I}^{\beta} x_i^{\otimes w_i} \right)}{\partial \beta} = \begin{cases} 1 & \text{if } 0 \leq \bigotimes_{i \in I}^{\beta} x_i^{\otimes w_i} \leq 1, \\ 0 & \text{otherwise,} \end{cases} \quad (103)$$

$$\frac{\partial \left( \bigotimes_{i \in I}^{\beta} x_i^{\otimes w_i} \right)}{\partial w_i} = \begin{cases} (x_i - 1) & \text{if } 0 \leq \bigotimes_{i \in I}^{\beta} x_i^{\otimes w_i} \leq 1, \\ 0 & \text{otherwise.} \end{cases} \quad (104)$$

Note that for a True input  $x_i = 1$  the corresponding gradient for the output in terms of  $w_i$  is  $(x_i - 1) = 0$ , which means that it provides no gradients for updating  $w_i$  even when no upper bound clamping is applied. The bias  $\beta$  will have to be adjusted instead to obtain the desired output, such as a False output when all inputs are True.

#### G.3.2 Gradient-transparent clamping

Note that the gradients are non-zero only inside the real-unit interval, so any clamping that is applied normally nullifies the gradient information so that learning receives no useful gradient information. This is problematic given that a major part of the operating range of the logical conjunction undergoes clamping, depending on the parameters. The solution to this problem is to only perform value clamping but leave the gradients untouched, through gradient-transparent clamping. In this case the output gradient for  $\beta$  is always 1 and for  $w_i$  it is always  $(x_i - 1)$ , both inside and outside the clamped region.

The parameterization of the proposed logical neurons is especially well-behaved and bounded, with  $0 \leq w_i \leq 1$  and  $0 \leq \beta \leq 1 + \sum_i w_i$  normally applying without loss of functionality. This is because any parameter configuration from these constraints can allow the useful functional range of bounded ReLU to be accessed. Consequently, the learning optimizer can set robust update step sizes and limit parameter updates, which means that the presence of transparent gradients from parameters in clamped situations requiring larger update steps can be handled feasibly.

The contradiction loss  $\sum_j \max(0, L_j(\beta, W) - U_j(\beta, W))$  also involves clamping with the intent that a loss term should be activated only when the lower bound  $L_j$  is higher than upper bound  $U_j$ . If there are no contradictions, we would not want to make any parameter updates based on this loss. So the clamping can be performed such that the gradients are also clamped and a zero gradient set when there are no contradictions. This means normal clamping can be performed, especially for loss terms in cases where no learning should be performed.

#### G.3.3 Downward pass

Downward pass at a weighted Łukasiewicz conjunction can be determined by firstly unclamping its output, which involves changing a clamped lower bound at  $L_{\otimes} = 0$  to the minimum unclamped value  $L_{\otimes} = \beta - \sum_{i \in I} w_i$  and similarly changing a clamped upper bound at  $U_{\otimes} = 1$  to the maximum

unclamped value  $U_{\otimes} = \beta$ . Unclamping and extending the bounds where necessary ensures that all downward pass explanations can be generated.

$$\bigotimes_{i \in I}^{\beta'} x_i^{\otimes w_i} = \max(0, \min(1, \beta - \sum_{i \in I} w_i(1 - x_i))) \quad (105)$$

$$= \max\left(0, \min\left(1, \bigotimes_{i \in I}^{\beta} x_i^{\otimes w_i}\right)\right). \quad (106)$$

Unclamping  $x' = \min(x_{\max}, x)$  is simply  $\max(x', x)$  and similarly for  $x' = \max(x_{\min}, x)$  we have  $\min(x', x)$  as unclamping operation. Here the  $x$  values are recalculated with gradient-tracking, although during clamping they are detached in the subtraction. Note that disabling gradient-tracking is not required for unclamping as it was for clamping, since there is no subtraction or its destructive interference that can nullify gradients. The unclamping of the bounded output to its extremes can then continue as

$$L_{\otimes} = \min\left(\bigotimes_{i \in I}^{\beta'} x_i^{\otimes w_i}, \beta - \sum_{i \in I} w_i\right), \quad (107)$$

$$U_{\otimes} = \max\left(\bigotimes_{i \in I}^{\beta'} x_i^{\otimes w_i}, \beta\right). \quad (108)$$

The functional inverse can then be determined with known values for the conjunction output and all but one of the inputs  $i$ , where backward inference can then calculate a value for the remaining subject input  $j$ . The calculated proof for the target input is then also clamped to the real-unit interval before proof aggregation is performed.

$$\bigotimes_{i \in I}^{\beta} x_i^{\otimes w_i} = \beta - \sum_{i \in I} w_i(1 - x_i) \quad (109)$$

$$w_j(1 - x_j) = \beta - \sum_{i \in I \setminus j} w_i(1 - x_i) - \bigotimes_{i \in I}^{\beta} x_i^{\otimes w_i} \quad (110)$$

$$(1 - x_j) = \beta/w_j - \sum_{i \in I \setminus j} w_i(1 - x_i)/w_j - \left(\bigotimes_{i \in I}^{\beta} x_i^{\otimes w_i}\right)/w_j \quad (111)$$

$$x_j = \max\left(0, \min\left(1, 1 - \beta/w_j + \sum_{i \in I \setminus j} w_i(1 - x_i)/w_j + \left(\bigotimes_{i \in I}^{\beta} x_i^{\otimes w_i}\right)/w_j\right)\right). \quad (112)$$

The conjunctive syllogism  $(p \otimes \neg(p \otimes q)) \rightarrow \neg q$  provides logical inference to determine a backward inference result equivalent to the functional inverse since

$$x_j = 1 - \beta/w_j + \left(1 - \left(1 - \sum_{i \in I \setminus j} w_i(1 - x_i)\right)\right)/w_j + \left(\bigotimes_{i \in I}^{\beta} x_i^{\otimes w_i}\right)/w_j \quad (113)$$

$$= 1 - \beta/w_j + \left(1 - \bigotimes_{i \in I \setminus j}^1 x_i^{\otimes w_i}\right)/w_j + \left(\bigotimes_{i \in I}^{\beta} x_i^{\otimes w_i}\right)/w_j \quad (114)$$

$$= \beta/w_j \left(\left(\bigotimes_{i \in I \setminus j}^1 x_i^{\otimes w_i}\right)^{\otimes 1/w_j} \rightarrow \left(\bigotimes_{i \in I}^{\beta} x_i^{\otimes w_i}\right)^{\oplus 1/w_j}\right). \quad (115)$$

where the weighted Łukasiewicz implication is generally defined as

$$\begin{aligned} \beta(x^{\otimes w_x} \rightarrow y^{\oplus w_y}) &= \max(0, \min(1, 1 - \beta + w_x(1 - x) + w_y y)) \\ &= \beta((1 - x)^{\oplus w_x} \oplus y^{\oplus w_y}). \end{aligned} \quad (116)$$

Note that there is a negation of the partial conjunction which also involves a swapping of its lower and upper bounds in the backward inference calculation. The unclamped gradients obtained over the entire operating range with gradient-transparent clamping calculates as follows from 112

$$\frac{\partial x_j}{\partial \beta} = -1/w_j, \quad (117)$$

$$\frac{\partial x_j}{\partial w_i} = \frac{1 - x_i}{w_j}, \quad (118)$$

$$\frac{\partial x_j}{\partial w_j} = -\frac{\beta}{w_j^2} - \sum_{i \in I \setminus j} \frac{w_i(1 - x_i)}{w_j^2} - \frac{\bigotimes_{i \in I}^{\beta} x_i^{\otimes w_i}}{w_j^2} \frac{\partial \left(\bigotimes_{i \in I}^{\beta} x_i^{\otimes w_i}\right)}{\partial w_j}. \quad (119)$$

For weights  $0 \leq w_j \leq 1$  smaller than 1 all backward inference gradients  $\lim_{w_j \rightarrow 0} \partial x_j / \partial \beta = \infty$ ,

$\lim_{w_j \rightarrow 0} \partial x_j / \partial w_i = \infty$ , and  $\lim_{w_j \rightarrow 0} \partial x_j / \partial w_j = \infty$  tend to become large as the weights become smaller.

Gradient clipping can deal with these large gradients when performing learning updates, or reverting to clamping gradients as per usual can also be considered.

### G.3.4 Gradient supplplantation

The introduction of a threshold-of-truth  $\alpha$  allows for non-zero gradients in the regions where clamping will normally be performed. A tailored piecewise linear activation function provides a range of  $[0, 1 - \alpha]$  to express the clamped `False` domain of a logical conjunction, thus it has a positive non-zero gradient associated with the value-region of classical `False`. The gradient states that a positive change in the function input will result in a positive increase in the output truth value, even though an insufficient step-size could still result in a classical `False` output. Adjusting the threshold-of-truth  $2/3 \leq \alpha \leq 1$  can change the gradient magnitude in the classical regions, so that a more conservative gradient can be obtained for these regions that could make learning updates more accurate.

In contrast, gradient-transparent clamping utilizes the existing gradients of the same magnitude as the unclamped region, so it offers more aggressive gradients in the classical regions compared to piecewise linear activations. An approach of gradient supplplantation in gradient-transparent clamping could ensure equivalent gradient magnitudes to piecewise linear activations, but without the need to support the classical symmetry in range  $[\alpha, 1]$  associated with `True`. Output values also do not have to be relaxed by  $\alpha$  so that `False` is still only at an output of 0, but arbitrary gradients can be provided in the clamped regions with gradient-transparent clamping.

Basic gradient supplplantation alters gradient-tracked values by scaling their gradients with a provided scalar  $a$  under specified conditions. In gradient-transparent clamping the addition of gradient supplplantation can scale the effective gradient where value clamping has been applied. Bounded ReLU can then exhibit an arbitrarily scaled gradient in its rectified region to allow for more accurate learning updates, since smaller gradients can be chosen for clamped regions. The scaling-based gradient supplplantation uses indicator or boolean condition functions readily available in automatic differentiation libraries, in addition to value detachment  $x^*$  from the computational graph for gradient-tracked  $x$ , and the adapted gradient-transparent functions are thus applied as

$$\min(x_{\max}, x)^{(a)} = x(x \leq x_{\max}) + (ax - ax^* + x_{\max})(x_{\max} < x), \quad (120)$$

$$\max(x_{\min}, x)^{(a)} = (ax - ax^* + x_{\min})(x < x_{\min}) + x(x_{\min} \leq x). \quad (121)$$

Normal clamping that involves setting gradients in the clamped regions to zero, would then correspond with zero-scaled gradient-transparent clamping  $\min(x_{\max}, x)^{(0)}$  and  $\max(x_{\min}, x)^{(0)}$  that uses gradient supplplantation with a scaling factor of  $a = 0$ .

## H Empirical evaluation (continued)

### H.1 Smokers and friends

This section has further details about the **Smokers and friends** experiment, to augment the overview given in the main paper in **Empirical evaluation** section (Section 7) of the main paper, on page 7.

**Dataset.** We use the Smokers and friends dataset from LTN experiment  $\mathcal{K}_{\text{exp2}}$  [19] with the small universe of constants a-h. Initial facts on three predicates, namely smokes  $S(x)$ , cancer  $C(x)$ , and friends  $F(x, y)$  (open-world) are given.

**Axioms.** We use the logical formulae, consisting of 5 plausible axioms, of the Smokers and Friends experiment in LTN experiment  $\mathcal{K}_{\text{exp2}}$  [19]. We repeat the experiments including axioms induced by MLN [15] (total 8 axioms) on this dataset, seeded with the original 5 axioms. In particular we learn MLN formula structure and associated weights on this data, using *Alchemy 2* [16] with the command `./learnstruct -i smoking.mln -t smoking-a-h.db -o smoking-a-h-out.mln` (please find the attached code in the supplementary files).

**Experiments.** We conduct four experiments, two on the smaller 5 axioms, and another two on the larger 8 axioms for the same data. The learning parameterization of the experiments in Table 5 show ablation of the logical connective weights to ascertain its effect. The objective of the experiment is formulated in the loss function as  $(1 + \text{contradiction}) / (1 + \text{factalign} + \text{tightbounds})$ , where the goal is to reduce contradiction over the entire network as much as possible by balancing relaxation (through learning) of initial facts and all inferences made. This means that initial fact bounds and axiom bounds form learnable parameters and are updated through gradient descent with the loss gradient w.r.t. each parameter. A larger axiom set will place more simultaneous constraints that could produce more inconsistencies.